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Software Engineering

Assignment
→ Read T&C while installing an application

I.E.E.E (SRS) template
software requirement specification

The project structure is as follows:

(1) Introduction

Software

(1) Instructions or computer programs that when executed provide desired features, functionalities & performance.

(2) Data structures that enable the programs to adequately manipulate information.

(3) Documents that describe the operation & use of the programs.

Note: 2 types of documents → for user
→ for developer

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Evolutionary role of software

(1) **Product**: As a product it delivers the computing potential embodied by computer hardware or more broadly a N/W of computers that are accessible by local or global hardware.

(2) **Vehicle or Platform**: As a vehicle used to deliver the product, software acts as the basis for control of the computer → (1), communication of information → (2) & the creation and control of other programs. → (3)

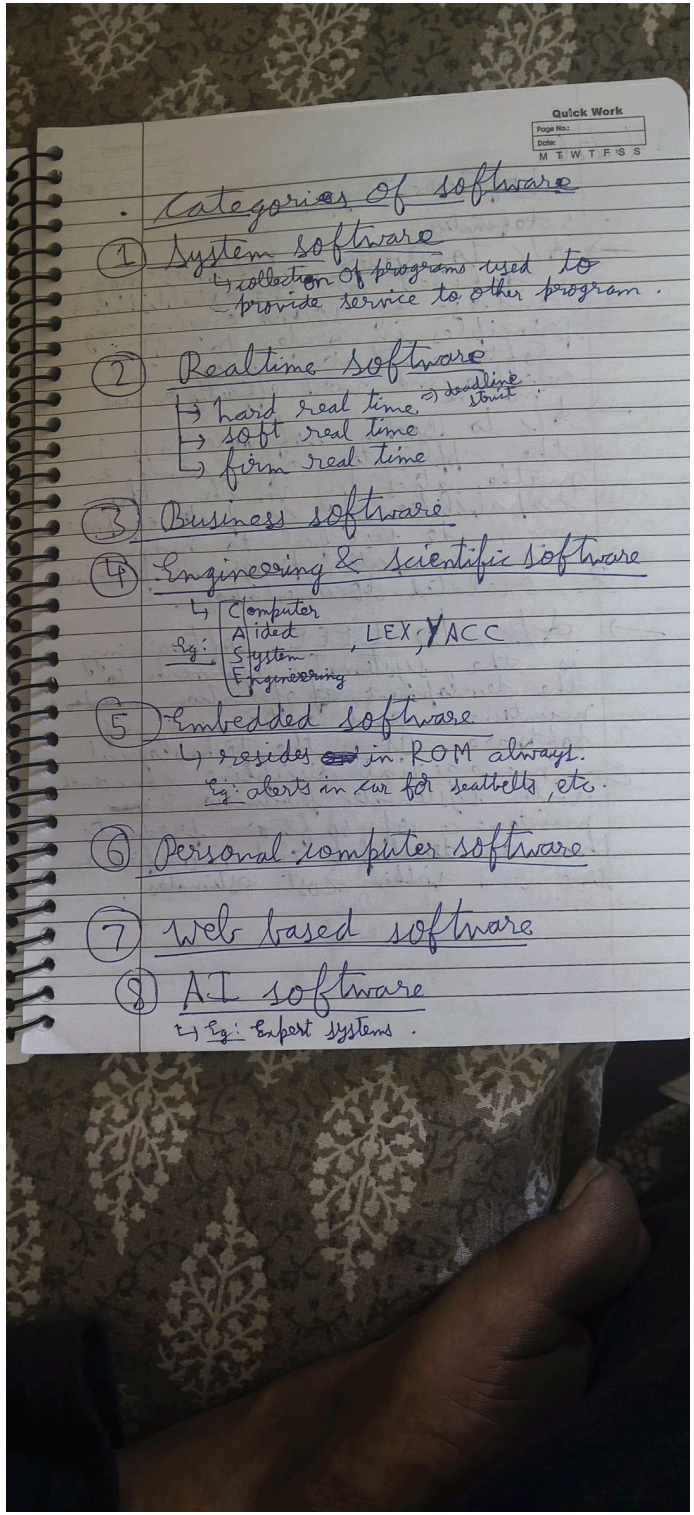
Functions of software

(1) It delivers the most important product which is information & data.
→ processed data.

(2) It provides the gateway for worldwide information.

(3) It provides the means of exchanging information.

(4) It manages the Business information


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Categories of software
1 System software
↳ collection of programs used to provide service to other program.
2 Realtime software
↳ hard real time ⇒ deadline strict.
↳ soft real time
↳ firm real time
3 Business software
4 Engineering & scientific software
↳ Eg: Computer Aided System Engineering, LEX, YACC
5 Embedded software
↳ resides in ROM always.
Eg: alerts in car for seatbelts, etc.
6 Personal computer software
7 Web based software
8 AI software
↳ Eg: Expert systems.

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17/08/26

Software Engg.

- 3 definitions

→ A/c to Bauer, Software Engg. is defined as the establishment and use of sound engineering principles in order to obtain software that is economical, reliable and work efficiently on real machine.

→ A/c to Boehm, software Engg. is the application of science & mathematics by which the capabilities of computer equipment are made useful to man via computer programs, procedures & related documentation.

→ A/c to IEEE, software engg. is the systematic approach to the development, operation, maintainence and retirement of software. It is the technological & managerial discipline concerned with systematic production & maintainence of software products that are developed and modified on time and within cost estimates.

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Goals of SWE

1. To improve quality of S/W.
2. To improve productivity.
3. To improve the job satisfaction of software Engineers.

Important terms

→ Project size ⇒ always expressed in LOC (lines of code)

⇒ It is a major factor that determines the level of management control & types of tools & techniques required for a software project.

⇒ categories of project size

1. trivial: *involves one project almost always working part-time. Time spent is few days to few weeks.
   * Program contains around 500 LOC
   * S/W would contain 10-20 subroutines.
   * They are mostly personal S/W for exclusive use of programmers.
   * Very little need for formal analysis, elaborate design documentation or extensive planning.

(modules = routines)

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dedicated

(2) Small: * One programmer
* 1-6 months
* 1000 - 2000 LOC
* 25-50 routines
* usually do not have interaction among programs.
* Generally used for small commercial applications including report generation & simple data manipulation
* Negligible interaction b/w customer & programmer

(3) Medium: * 2-5 programmers
* 1-2 years
* 10,000 to 50,000 LOC
* 250-1000 routines
* Medium interaction b/w programmers & customers.
* A certain degree of formality is required in planning & documentation & project reviews.

Eg: compilers, MIS, portals, etc.

(4) Large: * 5-20 programmers.
* 2-3 years
* 50,000 to 1 lakh LOC
* Entire project is combination of several subsystems.
* A significant interaction b/w software programs.

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* It requires multi-level management.
* It involves extensive planning analysis, risk estimation, standardized documentation & formal reviews.
Eg: Small time sharing systems.

(5) Very large: * 100-1000 programmers.
* 4-5 years
* up to 10 lakh LOC
* The subsystems are very complex & are developed seperately under expert guidance.
* Prominent features are multi-tasking, real time processing & tele-communication.
Eg: Large OS (developed for mainframe, supercomputers, etc.)

(6) Extremely large: * 2000-5000 programmers.
* up to 10 years of development.
* 1 million to 10 million LOC.
* The entire project is an integration of several very large systems which further consists of several large subsystems.
* distributed systems, & multi-tasking, real time processing & tele-communication are prominent features.

Eg: hard system tasks like missile launch systems.